

Unit 8, Lesson 1: Identifying Key Features of Quadratic Functions



Benchmark

MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function, and determine and interpret its key features.



Learning Target

- ☐ I can identify key features of a quadratic given a graph, a table, or an equation.



8.1.1 Warm-Up: Food Stylist

Kim Kissling changed careers from working a corporate job to working as a food stylist after learning about the job during a career day at cooking school. As a food stylist, Kim takes photographs for cookbooks, magazines, advertising campaigns, product packaging, and television advertisements.



1. What do you think would be interesting about working as a food stylist?
2. Kim said two important skills for her job are being super organized and not wasting time. How can you use these skills as a student?



8.1.2 Guided Instruction: Key Features of Quadratics

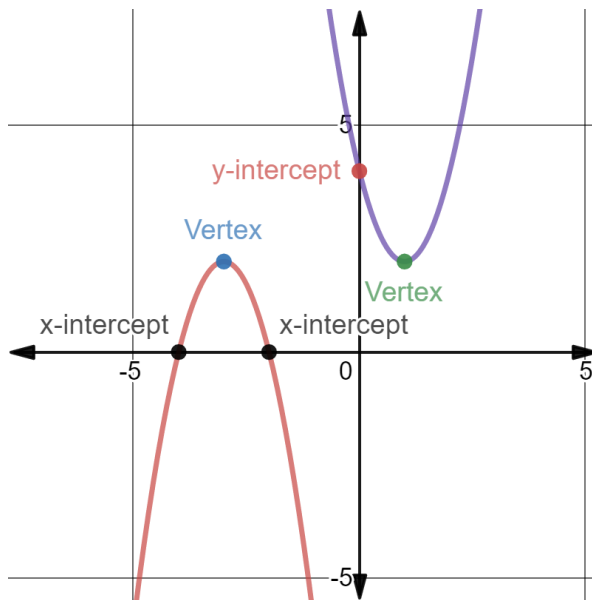
The graph of a quadratic equation is called a parabola.

The **vertex** of a parabola is the point where the parabola changes direction.



The **intercept** is the value of a variable when all other variables in the equation equal 0. On a graph, the intercept is the value where a function crosses an axis.

1. Work with your partner to describe the red and purple parabolas. Note any similarities or differences.



2. On the graph circle where the red parabola has negative y -values.
3. Draw a square on the graph where the red parabola has positive y -values.
4. Discuss with your partner what key feature can be helpful in determining where a quadratic function has positive y -values and negative y -values.
5. Discuss with your partner why the purple parabola only has positive y -values.



A **line of symmetry** is a line that divides a figure into two congruent parts, so that the reflection of either part across the line maps precisely onto the other.

6. Every parabola has a line of symmetry. Draw the line of symmetry for the red and purple parabolas.
7. What do you notice about the two lines of symmetry?



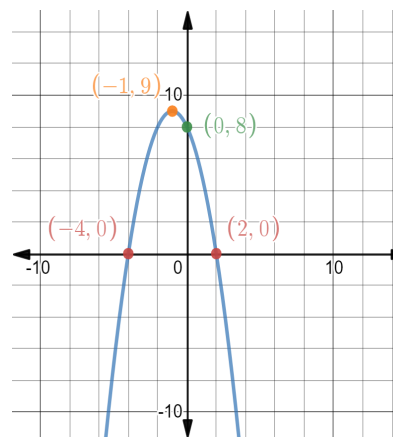
8.1.3 Collaboration: Key Features of Quadratics Using Equations

One of the key features studied with lines is whether the function is increasing or decreasing. Unless a line has a slope of 0 or an undefined slope, the line is always increasing or decreasing as x increases (from left to right).

1. With your partner, use the red and purple parabolas to make a conjecture about where parabolas are increasing and decreasing.

The table shows five forms of the same quadratic equation, $y = -1(x + 1)^2 + 9$, the graph of the quadratic, and a list of key features. The key features are color coded in each of the representations.

$$\begin{aligned}y &= -1(x - (-1))^2 + 9 \\y &= -1(x + 1)^2 + 9 \\y &= -1(x - (-4))(x - 2) \\y &= -1(x + 4)(x - 2) \\y &= -1x^2 - 2x + 8\end{aligned}$$



Name	Key Feature
x -intercept	$(-4, 0)$
x -intercept	$(2, 0)$
y -intercept	$(0, 8)$
vertex	$(-1, 9)$
axis of symmetry	$x = -1$
increasing	$x < -1$
decreasing	$x > -1$
positive	$-4 < x < 2$
negative	$x < -4$ or $x > 2$

2. The equations $y = -1(x - (-1))^2 + 9$ and $y = -1(x + 1)^2 + 9$ are very similar. How can the vertex be determined when the equation $y = -1(x + 1)^2 + 9$ is given?

3. The equations $y = -1(x + 1)^2 + 9$ and $y = -1(x + 4)(x - 2)$ both represent the same quadratic. Rewrite both so that they are in the form $y = ax^2 + bx + c$. Show your work.

$y = -1(x + 1)^2 + 9$	$y = -1(x + 4)(x - 2)$

4. The equations $y = -1(x - (-4))(x - 2)$ and $y = -1(x + 4)(x - 2)$ are equivalent. How can both x -intercepts be determined if the equation $y = -1(x + 4)(x - 2)$ is given?

5. The equation $y = -1x^2 - 2x + 8$ can be used to determine the y -intercept. Show how each of the three forms can be used to find the y -intercept by substituting 0 for x .

$y = -1x^2 - 2x + 8$	$y = -1(x + 1)^2 + 9$	$y = -1(x + 4)(x - 2)$

6. A table of values for $y = -1x^2 - 2x + 8$ is shown.

- a. Check all the key features that can be found using the table of values.

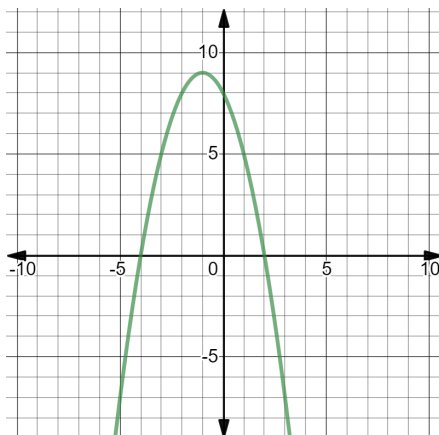
- ☐ x -intercepts
- ☐ y -intercept
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

x	y
-3	5
-2	8
-1	9
0	8
1	5
2	0
3	-7

- b. Discuss with your partner the key features that were not found, but that could be found if the table of values included more values.

7. A graph of the function $y = -1(x - 2)(x + 4)$ is shown. Check all the key features that can be found using the graph.

- ☐ x -intercepts
- ☐ y -intercept
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

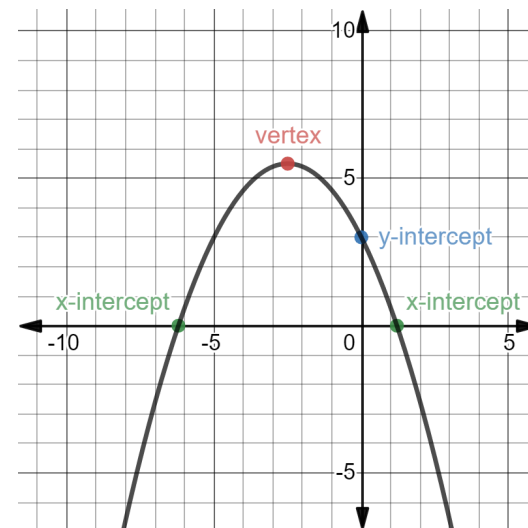


8. A table of values for $y = -\frac{2}{5}x^2 - 2x + 3$ is shown. Check all the key features that can be found using the table of values.

- ☐ x -intercepts
- ☐ y -intercept
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

x	y
-3	5.4
-2	5.4
-1	4.6
0	3
1	0.6
2	-2.6
3	-6.6

9. The equation $y = -\frac{2}{5}x^2 - 2x + 3$ is shown on the coordinate grid. Which, if any, of the ordered pairs of the marked key features can be determined from the graph?



Determining all of the key features of a quadratic may require using a graph, different forms of an equation, and a table of values.



A **quadratic function** is a polynomial function with degree of 2.



8.1.4 Guided Practice: Key Features of Quadratics Using Equations

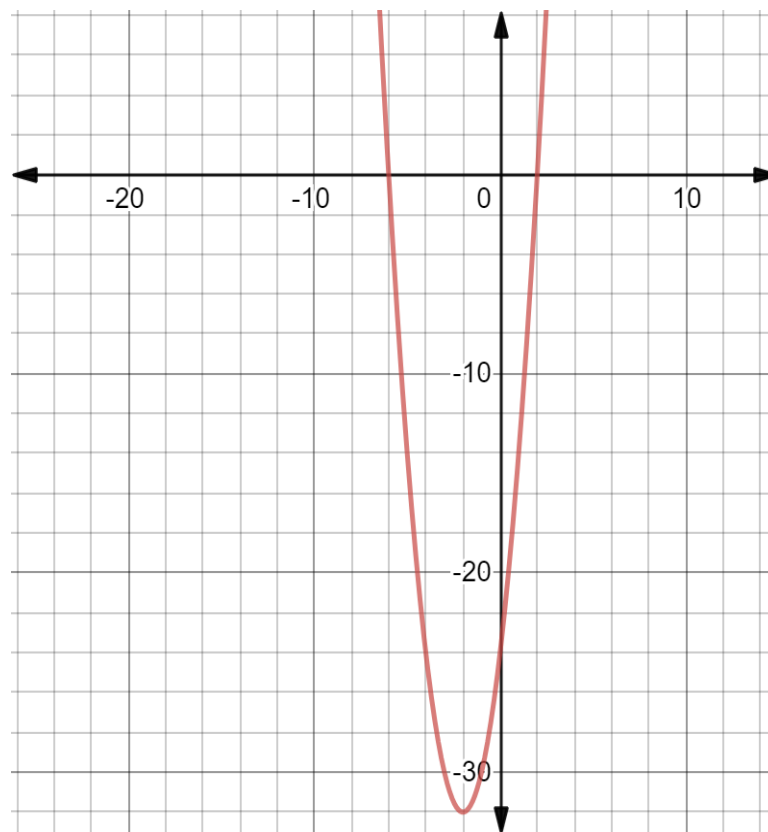
Three different equations for a quadratic function are given.

- Use the equations to determine the key features. If an equation cannot be used to determine a key feature write "no equation."

Standard Form	Vertex Form	Factored Form
$y = 2x^2 + 8x - 24$	$y = 2(x + 2)^2 - 32$	$y = 2(x + 6)(x - 2)$

Name	Form Used	Key Feature
x-intercepts		(____, ____) (____, ____)
y-intercept		(____, ____)
vertex		(____, ____)
axis of symmetry		$x = \underline{\hspace{2cm}}$
increasing/ decreasing		Increasing {x: _____} Decreasing {x: _____}
positive/negative		Positive {x: _____} Negative {x: _____}

- Label the key features on the graph of $y = 2x^2 + 8x - 24$. Then, if possible, circle the key feature in the table of values.

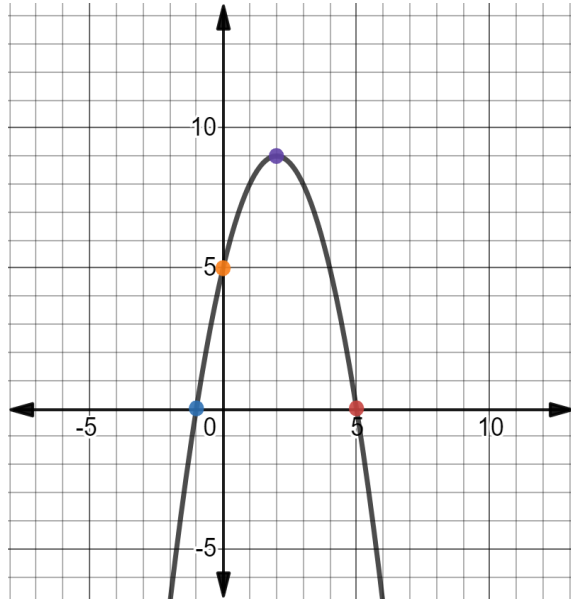


x	-3	-2	-1	0	1	2	3
y	-30	-32	-30	-24	-14	0	18



8.1.5 Wrap-Up: Cool Down

1. The graph of a quadratic equation is shown. Label the four colored points on the parabola with the ordered pair.
2. Use words, ordered pairs, or symbols to describe the location of each key feature on the given parabola.



Key Feature	Location of Key Feature
x -intercepts	
y -intercept	
vertex	
axis of symmetry	
increasing/decreasing	
positive/negative	



Unit 8, Lesson 2: More Key Features of Quadratic Functions



Benchmark

MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function, and determine and interpret its key features.



Learning Targets

- ☐ I can identify key features of a quadratic given a graph, a table, or an equation.
- ☐ I can graph a parabola given an equation or a table.